

Editorial

by Emi Tanaka

In this issue

On behalf of the Editorial Board, I am pleased to present Volume 18, Issue 3 of the R Journal. This issue features twenty research articles spanning a wide range of developments in statistical methodology, machine learning, time series, spatial analysis, causal inference, visualisation, and scientific computing.

Several articles introduce new tools for computationally challenging statistical problems. **hdtg** provides methods for simulating from high-dimensional truncated multivariate normal distributions, while **ODRF** develops an implementation of oblique decision trees and random forests. **psvmSDR** provides tools for sufficient dimension reduction using principal machines, and **DRIP** brings jump regression methods to image analysis. **SelectionBias** implements a range of methods for bounding selection bias in causal estimands. **LBBNN** provides a scalable approach to sparse and interpretable Bayesian neural networks, making use of variational inference and GPU acceleration. The issue also includes **rSRD** supporting the sum of ranking differences procedure, and **xplainfi** brings together a broad range of feature importance methods with tools for statistical inference.

Time series modelling is also well represented in this issue. **mtarm** provides Bayesian estimation and forecasting for multivariate threshold autoregressive models, while **echos** introduces automated time series forecasting using echo state networks. **MSTest** provides a collection of procedures for testing the number of regimes in Markov-switching models, **VARshrink** brings together shrinkage methods for high-dimensional vector autoregressive models, and **iAR** provides tools for modelling irregularly observed time series. **demofit** brings parametric and stochastic approaches to mortality modelling and forecasting into a unified framework.

There are also several contributions in spatial, spatiotemporal, and multivariate analysis. **ldmppr** provides tools for modelling marked point processes where the marks may depend on location, while **sugarglider** extends glyph-map visualisation for exploring spatiotemporal data. **goSorensen** provides methods for comparing gene lists using Gene Ontology information, **admix** supports estimation, testing and clustering for admixture models, and **dbrobust** provides tools for robust distance-based analysis and visualisation of mixed-type data.

Finally, **metafrontier** provides a comprehensive set of tools for metafrontier analysis of efficiency and productivity.

All packages discussed are available on CRAN or Bioconductor. Supplementary material with fully reproducible code is available for download from the R Journal website.

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